

1 Current Status Information on the Operation

This document describes the current status information, actual data and KPIs pertaining to an operation. Basically, this data results from the operation-related data collection in BDE (Shop Floor Data Collection). The document [data structure of operation](#) describes the operation data.

The below-described data is available in some applications. The categories used there might deviate from the categories mentioned below. It is not a fault if a piece of information is not available in a report/evaluation.

Status

Status (LED), status text

The status represents the current condition of an operation. Possible statuses are:

 prepared:	the operation has not yet been started
 running	the operation is currently being logged on
 automatically interrupted:	the operation is currently being logged on but has been interrupted automatically at the end of the shift due to the shift automatic.
 interrupted	the operation has been interrupted.
 finished:	the operation has been logged off.

Status since

Time since the order status is available.

Predecessor status

Status of the predecessor operation. This status indicates whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.



The predecessor status only affects one order. The predecessor status of the first operation of an order is always "finished", no matter if there is a preceding operation in another order due to a relationship (order network).

Secondary status

Displays the currently set secondary status.

See [here](#) to change the secondary status.

Please note: You can configure secondary statuses for specific projects while customizing HYDRA.

Resource status

By default, the resource status refers to "person", "tool", "material".

See [here](#) to change the resource status.

Target times

The document entitled [data structure of operation](#) describes the target times. Further data:

Total target setup time

Target setup time + dynamic setup time + target retooling time

Execution time

Total target setup time + processing time

Target labor utilization

The target labor utilization refers to the workforce requirements planned for the processing of the operation. Only relevant with BDE-AEV, otherwise the field is 0. As part of a customization, you can have the field completed via the interface.

Buffer time

HYDRA [Scheduling](#), if activated, calculates the buffer time.

The system determines the buffer time from the difference between the latest start date (LST) and the earliest start date (EST) for an operation.

Please note: The system stores the total of buffer times of all operations in the [Order](#) (header) in the field *OP buffer*.

Reducible time

HYDRA [Scheduling](#), if activated, calculates the reducible time.

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available (basic end date will be exceeded), then HYDRA will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the wait times and the transport time.

This value indicates how many (more) hours can be reduced from the lead time of an order. This time results from the sum of the:

- difference from the current waiting time and the minimum waiting and the
- difference from the current transport time and the minimum transport time

Reducible time = (current waiting time - minimum waiting time) + (current transport time - minimum transport time). These differences are displayed here as totals.

The document entitled bde-bk.doc describes the configuration of reduction strategies in the section *reduction strategies*.

You can find further information in the section *Reduction* of the Glossary (bde-glos.doc).

Target production time per piece (time/piece)

The time per piece is the calculated target duration per 1 or 1000 produced units of the corresponding quantity unit (primary). The calculation includes the cycle time and partitioning of the operation.

Example: The quantity unit is pieces. The time per piece is calculated for min/piece and min/1000 pieces. Calculation is as follows:

min/piece for the unit (cycle time (sec/1000) / 1000 / 60) * partitioning

min/1000 pieces: for the unit (cycle time (sec/1000) / 60) * partitioning

Actual times

Updating of actual times depends on the workplace. When it comes to an individual workplace assigned to a terminal with operation mode "MDE", times are updated cyclically approximately every two minutes or when the machine/workplace status is changed the next time. In case of a group workplace or an individual workplace that is not assigned to a terminal or to a terminal with operation mode "BDE", times are only updated when the operation is interrupted or logged off.

Setup time

Times posted onto the resource performance account 7 (SET)

Processing time

Times posted onto the resource performance account 11 (MUT)

Execution time

Setup time + processing time

Downtime

Sum total of the times posted to the resource performance accounts 1 - 6 and 8-10.

Occupancy time

Setup time + processing time + downtime

Lead time

The lead time results from the period of time between:

- the first logon of the operation and
- the last logoff of the operation (if the operation is finished) or
- the last interruption (if the operation is not finished).

Lead times are synchronized based on the Gregorian calendar (the shift calendar is not taken into account).

The lead time is 0 if no difference can be calculated (between logging on and off).

**Wait time **

The wait/idle time includes those times of the lead time during which the operation was not logged on.

Times are calculated based on the BDE log records. Lead times are synchronized based on the Gregorian calendar (the shift calendar is not taken into account).

The wait time is 0 if there is only one log record for an operation.



The total of wait times of all operations does not necessarily have to correspond to the order's wait time. This is particularly true if you produce operations simultaneously.

Transition time

The transition time of an operation refers to the time period between

- the (last) actual logoff of the current ("preceding") operation and
- the (first) actual logon of the next ("subsequent") operation. The transitional time is 0 if two OPs coincide.

Transition times are synchronized based on the Gregorian calendar (the shift calendar is not taken into account).

Please note: According to technical literature, the transition times often represent the largest share of the lead time. In contrast to this, the occupancy/assignment time often only amounts to approximately 10-20%.

Personnel deployment/labor utilization

Labor time posted onto the operation.

Actual labor utilization is updated when the person is logged off from the operation or the person logs off from the OP.

Actual production time per piece (time/piece)

The time per piece is the calculated actual duration per 1 or 1000 produced units of the corresponding quantity unit (primary). The calculation includes the cycle time and partitioning of the operation.

Example: The quantity unit is pieces. The time per piece is calculated for min/piece and min/1000 pieces. Calculation is as follows:

min/piece: for the unit (actual cycle (sec/1000) / 1000 / 60) * partitioning

min/1000 pieces: for the unit (actual cycle (sec/1000) / 60) * partitioning

Please note: You cannot calculate and view the actual time per piece for prepared operations.

Remaining run time

Remaining run time

Calculated remaining run time as per remaining run time formula 1.



The remaining run time is even calculated for finished operations - this applies for the RRT formula 1 and the RRT formula 2.

Remaining run time 2

Calculated remaining run time as per remaining run time formula 2.

KPIs

Target activity

The operation-related activity describes the quantity produced or to be produced per hour.

Target activity = Target quantity [primary quantity unit] / target processing time



The term "activity" is also used in another context (e.g. in HYDRA MDE). There, the term has another meaning (calculation rule).

Actual activity (gross)

Actual gross activity = Yield [primary quantity unit] / Assignment/occupancy time
= Yield [primary quantity unit] / Σ RPA 1 - 11

Actual activity (net)

Actual net activity = Yield [primary quantity unit] / RPA 11

Activity [%]

Activity [%] = Actual net activity / target activity

Target run time

The run time specifies the planned or required processing time in the format hrs.min:sec.

The target run time corresponds to the target processing time (see document [data structure of operation](#)).

Actual run time (gross)

Actual run time (gross) = Σ RPA 1 - 11

Actual run time (net)

Actual run time (net) = RPA 11

Run time [%]

Run time [%] = Actual run time (net) / target run time

Target remaining run time

Remaining run times are shown in the format hrs.min:sec. They can also be negative.

Target remaining run time = Target run time - actual run time (gross)



HYDRA Shop Floor Scheduling (HLS) uses the formula defined in the field "formula RRT 1" to calculate the processing time or remaining run time.

Remaining run time (gross)

Remaining run time (gross) = Actual assignment time / Yield * Quantity that is still to be produced

$$= \text{Actual assignment time} / \text{Yield} * (\text{Target quantity} - \text{Yield})$$

The quantities in primary quantity unit are used as target quantity and yield.

Remaining run time (net)

Remaining run time (gross) = Actual processing time / Yield * Quantity that is still to be produced

$$= \text{RPA 11 [sec]} / \text{Yield} * (\text{Target quantity} - \text{Yield})$$

The quantities in primary quantity unit are used as target quantity and yield.

Yield/target quantity [%]

Yield/target quantity [%] = 100 / Target quantity * Yield

The quantities in primary quantity unit are used.

Setup [%]

Setup [%] = 100 / Target setup time (total) * Actual setup time

$$= 100 / \text{Target setup time} + \text{Additional setup time} + \text{Target retooling time} * \text{RPA 7}$$

Processing [%]

Processing [%] = 100 / Target processing time * Actual processing time

$$= 100 / \text{Target processing time} * \text{RPA 11}$$

Labor utilization [%]

Labor utilization [%] = 100 / Target labor utilization * Actual labor utilization



The target labor utilization can be transferred according to the customer's requirements via the ERP interface.

Utilization efficiency (rate of capacity utilization)

Rate of capacity utilization = 100 / Actual assignment/occupancy time * Actual processing time

$$= 100 / \sum (\text{RPA 1} - \text{11}) * \text{RPA 11}$$

Throughput rate

Produced quantity/lead time (DLZ)

Formula: yield.primary / DLZAG

DLZAG= (operation logoff – operation logon (first logon))

Use the formula **trp_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Production process ratio

Main production time/lead time (DLZ)

Formula: $\text{rpa11} / \text{DLZAG} * 100$

$\text{DLZAG} = (\text{operation logoff} - \text{operation logon (first logon)})$

Use the formula **ppr_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Actual to planned scrap ratio

Scrap quantity/planned scrap quantity

Formula: $\text{scrap.primary} / \text{operation.plan.scrap.primary} * 100$

Use the formula **psc_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Scrap ratio

Scrap quantity/ produced quantity

Formula: $\text{scrap.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **scr_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Rework ratio

Rework quantity / produced quantity

Formula: $\text{rework.primary} / (\text{yield.primary} + \text{scrap.primary} + \text{rework.primary} + \text{problem.primary}) * 100$

Use the formula **rew_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.

Allocation ratio

Assignment time/lead time

Formula: $\text{BLZ} / \text{DLZAG} * 100$

$\text{BLZ} = \text{total occupancy} / \text{assignment time} = \text{BMK1-11}$

$\text{DLZAG} = (\text{operation logoff} - \text{operation logon (first logon)})$

Use the formula **all_ag** to customize the calculation. If the formula management does not include this formula, you have to create the formula in order to change the calculation.



You cannot change the formula components DLZAU and/or DLZAG.



The KPIs *throughput rate*, *production process ratio*, *actual to planned scrap ratio*, *scrap rate*, *rework ratio* and *allocation ratio* are only available if you enable the upgrade. [orst82](#).

RPA

SUT...BKS

Order-related times posted onto the relevant resource performance account.

Personnel deployment/labor utilization

Labor times posted onto the operation.

Total duration

Sum total of the times posted to the resource performance accounts 1 - 11.

Postings

First login

Point in time when the operation was first logged on.

Last login

Point in time when the operation was last logged on.

Last interruption

Point in time when the operation was last interrupted.

Last logoff

Point in time when the operation was last logged off.

Further data

This paragraph is not restricted to a special category of MOC applications.

Actual cycle

The system calculates the operation-related actual cycle as follows:

- 1) OP prepared: by definition the actual cycle is 0.
- 2) OP logged on: the actual cycle is the current actual cycle from the MDE machine connection. The application [Workplaces/machines](#) (transaction code wpov) also shows the actual cycle.
- 3) OP interrupted or logged off/finished: The system calculates an average cycle time. To calculate the average cycle time, divide the main production time posted onto the operation (RPA 11) by the total number of cycles posted onto the operation. The system only calculates the average cycle time for these order postings (no recalculation, e.g. after changes in the maintenance of postings dialog).

Please note for order-related KPIs

When calculating order-related KPIs (identified with the icon  in the document), the system integrates those operations matching the below-mentioned criteria:

- Data can be entered for the operation (flag at the operation)
- It is a "normal" operation or a split master (individual splits are not taken into account).
- The operation pertains to an active sequence.
- The operation is neither inactive nor deleted (control indicators <> "Y" and <> "D").

Merged operations are not taken into account.



A program integrated in the HYDRA Scheduler calculates order-related KPIs (identified by ) at cyclic intervals (by default: once a day). Further information on how to activate this program can be found [here](#).